

EQUIVARIANT UNIRATIONALITY OF CUBIC THREEFOLDS

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ABSTRACT. Using the recent proof of the Cassels–Swinnerton-Dyer conjecture for cubic surfaces, we finish the classification of G -unirational complex cubic threefolds. In particular, we prove that the Klein cubic threefold is $\mathrm{PSL}_2(\mathbb{F}_{11})$ -unirational, establishing that the essential dimension of $\mathrm{PSL}_2(\mathbb{F}_{11})$ is 3. This disproves a conjecture of Dolgachev that the essential dimension of a group is at least its Cremona dimension.

1. INTRODUCTION

Let X be a variety over a field k with a regular action of a finite group G . We say X is G -*unirational* if there exists a dominant G -equivariant rational map $V \dashrightarrow X$ where V is a finite-dimensional linear representation of G .

An important application of G -unirationality is for *essential dimension* (see e.g. [Rei10]). In fact, we can *define* the essential dimension of a finite group G over k , denoted $\mathrm{ed}_k(G)$, as the minimal dimension of a G -unirational variety over k with a faithful action of G .

Using Prokhorov’s classification [Pro12] of finite simple groups acting on rationally connected threefolds over $\mathbb{C}$, the finite simple groups G such that $\mathrm{ed}_{\mathbb{C}}(G) \leq 3$ were partially classified by Duncan [Dun13, Dun10], and Beauville [Bea14]. Namely, the groups of essential dimension ≤ 3 must be among the cyclic groups and the nonabelian groups

$$\mathfrak{A}_5, \quad \mathrm{PSL}_2(\mathbb{F}_7), \quad \mathfrak{A}_6, \quad \mathrm{PSL}_2(\mathbb{F}_{11}),$$

where $\mathrm{PSL}_2(\mathbb{F}_{11})$ was the only case not proved to have $\mathrm{ed}_{\mathbb{C}}(G) \leq 3$.

Prokhorov’s classification showed that there were exactly two unirational threefolds with a $\mathrm{PSL}_2(\mathbb{F}_{11})$ -action up to equivariant birational equivalence. Namely, there is the *Klein cubic threefold* V_3 , defined by the equation

$$x_1^2 x_2 + x_2^2 x_3 + x_3^2 x_4 + x_4^2 x_5 + x_5^2 x_1 = 0$$

in $\mathbb{P}^4$, and a smooth Fano threefold V_{14} with Picard rank 1 and anticanonical volume 14. Thus, the question of the essential dimension of $\mathrm{PSL}_2(\mathbb{F}_{11})$ is equivalent to determining whether these two threefolds are $\mathrm{PSL}_2(\mathbb{F}_{11})$ -unirational. Our main results are:

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Theorem 1.1. *The Klein cubic V_3 is $\mathrm{PSL}_2(\mathbb{F}_{11})$ -unirational, while V_{14} is not $\mathrm{PSL}_2(\mathbb{F}_{11})$ -unirational.*

Corollary 1.2. *We have $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11})) = 3$.*

Recall that the Cremona group of degree n , denoted $\mathrm{Cr}_n(k)$, is the group of birational automorphisms of $\mathbb{P}^n$. The *Cremona dimension* of a finite group G is the minimal degree n such that there is an embedding $G \hookrightarrow \mathrm{Cr}_n(k)$. In 2010, Igor Dolgachev conjectured that the Cremona dimension was always less than or equal to the essential dimension (see [DR15, §10] and [Dol25]). Since neither V_3 nor V_{14} is rational, the Cremona dimension of $\mathrm{PSL}_2(\mathbb{F}_{11})$ is 4. Thus, Theorem 1.1 also provides a counterexample to Dolgachev’s conjecture.

Our proof of Theorem 1.1 relies on the recent resolution of the Cassels–Swinnerton-Dyer conjecture for cubic surfaces:

Theorem 1.3 (Alexeev–Schreieder [AS26]). *Let X be an irreducible cubic surface over a field k . If X has an effective 0-cycle of degree coprime to 3, then X has a k -rational point.*

Note that the threefold case of the Cassels–Swinnerton-Dyer conjecture is still open, but we are still in a position to completely determine G -unirationality for cubic threefolds.

Definition 1.4. Let X be a G -variety over an algebraically closed field.

- X satisfies *Condition (A)* if, for every abelian subgroup A of G , the fixed point locus X^A is non-empty.
- X satisfies *Condition (A*)* if, for every abelian subgroup A of G , either X^A contains a rational curve or $X^{Z_G(A)}$ is nonempty, where $Z_G(A)$ is the centralizer of A in G .

Condition (A) and Condition (A*) are necessary for G -unirationality (see Lemma 2.4 and Corollary 4.2); in fact, they are even necessary for G -weak versality. While Condition (A*) is new, Condition (A) has been used for decades to study essential dimension [RY00]. For example, in [Dun16] it is shown that a faithful smooth del Pezzo G -surface S of degree ≥ 3 is G -unirational if and only if S satisfies Condition (A); in particular, this includes all smooth cubic surfaces. In [CTZ25], it is proved that Condition (A) implies G -unirationality of cubic threefolds, with the possible exception of four very specific families which were left unresolved. We see that Condition (A*) is sufficient to finish the remaining cases:

Theorem 1.5. *Let k be an algebraically closed field of characteristic 0. Let X be an irreducible cubic threefold over k , which is not a cone, with a faithful action of a finite group G . Then X is G -unirational if and only if it satisfies Condition (A*).*

Unlike Condition (A), Condition (A*) is *not* a G -equivariant birational invariant in general.

The structure of the paper is as follows. In §2, we review the bridge between arithmetic and equivariant geometry from [DR15] and discuss particular applications to cubic hypersurfaces. In §3, we demonstrate how the proof of the Cassels–Swinnerton-Dyer conjecture gives a G -unirationality construction for cubic threefolds that are equivariantly Pfaffian. In §4, we establish that Condition (A*) is necessary for G -unirationality by proving a generalization of *Going Down*. In §5, we prove that the Klein cubic is $\mathrm{PSL}_2(\mathbb{F}_{11})$ -unirational and establish Theorem 1.5. Finally, in §6, we show that V_{14} is not $\mathrm{PSL}_2(\mathbb{F}_{11})$ -unirational by showing that it does not satisfy Condition (A*).

2. PRELIMINARIES

Let k be a field of characteristic 0. Suppose X is an irreducible projective variety of finite type over k . Given a field extension K/k , we denote by X_K the K -variety $X \times_{\mathrm{Spec}(k)} \mathrm{Spec}(K)$. Let G be a finite group with a regular (not necessarily faithful) action of G on X . Given a (right) G -torsor $T \rightarrow \mathrm{Spec}(K)$, the *twist* of X by T is defined as the K -variety

$${}^T X := (T \times_K X_K)/G$$

where G acts diagonally.

Theorem 2.1 (Duncan–Reichstein [DR15]). *X is G -unirational if and only if, for every field extension K/k and every G -torsor $T \rightarrow \mathrm{Spec}(K)$, the twist ${}^T X$ is K -unirational.*

Definition 2.2. X is G -weakly versal if for every field K/k and every G -torsor $T \rightarrow \mathrm{Spec}(K)$, the twist ${}^T X$ has a K -rational point.

Observe that being G -weakly versal is a necessary condition for being G -unirational, since every unirational variety has a rational point. In particular, since twisting is functorial, if X has a G -fixed point, then it must be G -weakly versal.

Using this, we can see that Condition (A) is a G -equivariant birational invariant of smooth proper G -varieties in view of *Going Down*:

Theorem 2.3 (Kollár–Szabó [KS00]). *Suppose k is algebraically closed. Let A be an abelian group, and $X \dashrightarrow Y$ an A -equivariant rational map between two proper A -varieties over k . If X has a smooth A -fixed point, then Y has an A -fixed point.*

In particular, since every action on a linear representation fixes the origin, we get an easily checked necessary condition for G -weak versality (and thus also for G -unirationality).

Lemma 2.4. *Suppose X is a proper G -variety over an algebraically closed field k . If X is G -unirational, then X satisfies Condition (A).*

Cubic hypersurfaces. Now we specialize to the case of cubic hypersurfaces. We begin with a slight amplification of a well known result:

Proposition 2.5. *Suppose X is a k -variety such that there exists a field extension K/k such that X_K is an irreducible cubic hypersurface of dimension ≥ 2 that is not a cone over a lower-dimensional cubic. Suppose that L is a quadratic extension of k . If X_L has an L -rational point, then X has a k -rational point.*

Proof. Since X_K is a cubic hypersurface, we have an embedding of X into a Severi-Brauer variety P over k . Since X_K is a cubic hypersurface, the Brauer class of P must have period 3. Thus, P is projective space since X contains an L -point. Now, the L -point in X_L defines a unique line in P . Either the line is contained in X , which means there is a k -rational point, or it intersects X in one other point, which is the desired k -rational point. $\square$

The following is our main arithmetic input:

Theorem 2.6 (Kollár [Kol02]). *Let X be an irreducible cubic hypersurface of dimension ≥ 2 over a field k which is not a cone over a lower-dimensional cubic. Then X is k -unirational if and only if X has a k -rational point.*

From this and the theory above we have several ways to establish G -unirationality. The last two items in the following lemma appear to be new:

Lemma 2.7. *Let $X \subset \mathbb{P}^n$ be an irreducible cubic hypersurface of dimension ≥ 2 , which is not a cone, with an action of a finite group G . Suppose one of the following holds:*

- (a) X is G -weakly versal.
- (b) X contains a G -unirational subvariety Y .
- (c) X is H -unirational for some subgroup H of G with index 2.
- (d) X is $\tilde{G}$ -unirational where the $\tilde{G}$ -action arises from an exact sequence

$$1 \rightarrow C_2 \rightarrow \tilde{G} \rightarrow G \rightarrow 1.$$

- (e) k is algebraically closed and G is a finite subgroup of $\mathrm{PGL}_2(k)$.

Then X is G -unirational.

Proof. Let K/k be a field extension and $T \rightarrow \mathrm{Spec}(K)$ a G -torsor. Since the embedding $X \rightarrow \mathbb{P}^n$ comes from a rational multiple of the anticanonical line bundle, it is G -equivariant. Since twisting is functorial, we conclude that ${}^T X \subseteq {}^T \mathbb{P}^n$. If ${}^T X$ has a K -rational point, then so must ${}^T \mathbb{P}^n$; in this case, ${}^T X$ is again a cubic hypersurface in $\mathbb{P}^n$. In view of Theorem 2.1 and Theorem 2.6, to prove X is G -unirational, it suffices to prove that ${}^T X$ has a K -rational point.

Now, (a) is immediate. For (b), we recall that twisting is functorial, so $Y \hookrightarrow X$ gives rise to a morphism ${}^T Y \hookrightarrow {}^T X$. Since ${}^T Y$ has a K -rational point, so must ${}^T X$. Part (c) is essentially [Dun16, Theorem 3.2] and follows from the fact that if ${}^T X$ has an effective 0-cycle of degree 2, then it has a K -rational point.

For part (d), from the exact sequence of Galois cohomology

$$H^1(K, \tilde{G}) \rightarrow H^1(K, G) \rightarrow H^2(K, \mu_2) \simeq \text{Br}(K)[2],$$

we see that T lifts to a $\tilde{G}$ -torsor $\tilde{T} \rightarrow \text{Spec}(L)$ over a splitting field L of some element $\alpha \in \text{Br}(K)[2]$. By Merkurjev's theorem [Mer81, MS82], L can be obtained by adjoining square roots, i.e.,

$$L = K(\sqrt{\alpha_1}, \dots, \sqrt{\alpha_m}), \quad \text{for some } \alpha_1, \dots, \alpha_m \in K.$$

Our assumption implies that the cubic hypersurface $\tilde{T}X_L$ is L -unirational and thus has L -points. Note that

$$\tilde{T}(X_L) \simeq {}^T L(X_L) \simeq ({}^T X)_L.$$

Applying Proposition 2.5 m -times inductively, we conclude that ${}^T X$ has a K -rational point.

For part (e), let $F = k(\mathbb{P}^1/G)$, with the natural G -action on $\mathbb{P}^1$. Consider the G -torsor $S = \text{Spec}(k(\mathbb{P}^1)) \rightarrow \text{Spec}(F)$. Since k is algebraically closed, F is a C_1 -field by Tsen's theorem. Thus, the cubic ${}^S X$ has an F -point. Equivalently, there exists a G -equivariant rational map $\mathbb{P}^1 \dashrightarrow X$, which extends to a G -equivariant morphism

$$\varphi : \mathbb{P}^1 \rightarrow X.$$

Now, we have a K -morphism

$${}^T \varphi : {}^T(\mathbb{P}^1) \rightarrow {}^T X.$$

Recall that every form of $\mathbb{P}^1$ is a smooth plane conic. It follows that ${}^T(\mathbb{P}^1)$ has a rational point over a quadratic extension of K , and so does ${}^T X$. We conclude that ${}^T X$ has a K -rational point as above. $\square$

3. PFAFFIAN CUBIC THREEFOLDS

Here we follow [TZ24]. Let k be a field of characteristic 0. Let V be a k -vector space of dimension 6. Consider a 5-dimensional subspace

$$A \subseteq \Lambda^2(V^\vee),$$

and the subvariety

$$X_A := \{[a] \in \mathbb{P}(A) : \text{Pf}(a) = 0\}.$$

If a cubic threefold has such a description X_A , then we say that X has a *Pfaffian representation*.

Over an algebraically closed field, every smooth cubic threefold has a Pfaffian representation [Bea00], but this is not true in general. However, we have the following:

Theorem 3.1. *If a smooth cubic threefold X over k has a Pfaffian representation, then it has a rational k -point.*

Proof. From [Bea00, Proposition 8.2], a smooth Pfaffian cubic threefold contains a non-degenerate smooth curve of degree 5. Since k is infinite of characteristic 0, there is a smooth hyperplane section meeting the curve properly. This gives an effective 0-cycle of degree 5 on a smooth cubic surface. Thus, there exists a k -rational point by Theorem 1.3. $\square$

Remark 3.2. A main step in the proof of Theorem 1.3 in [AS26] is to show that a Pfaffian cubic surface always has a rational point, see [AS26, Theorem 1.4]. Theorem 3.1 also follows from that by taking appropriate hyperplane sections, and generalizes to Pfaffian cubic fourfolds.

Recall from [TZ24] that a faithful G -action on a cubic threefold X is G -equivariantly Pfaffian if the following occurs. There exists an extension

$$1 \rightarrow C_d \rightarrow \tilde{G} \rightarrow G \rightarrow 1$$

and a 6-dimensional representation V of $\tilde{G}$ such that the induced action on $\mathbb{P}(V)$ is G . Moreover, there exists a 5-dimensional subrepresentation $A \subseteq \Lambda^2(V^\vee)$ such that $X_A \cong X$ equivariantly.

Theorem 3.3. *Let X be a smooth cubic threefold with a faithful action of a finite group G . If X is G -equivariantly Pfaffian, then X is G -unirational.*

Proof. The 6-dimensional representation V of $\tilde{G}$ has dual exterior square containing a 5-dimensional representation. We know that $\tilde{G}$ corresponds to an element α of $H^2(G, k^\times)$ such that $6\alpha = 0$ and $5(-2\alpha) = 0$. Thus $2\alpha = 0$, and we may assume that $|\tilde{G}|/|G|$ is 1 or 2.

By Lemma 2.7(d), it then suffices to prove that X is $\tilde{G}$ -unirational. By Lemma 2.7(a), it suffices to show that X is $\tilde{G}$ -weakly versal. Thus, we consider a field extension L/k and a $\tilde{G}$ -torsor $T \rightarrow \operatorname{Spec}(L)$. The twist ${}^T X$ is still a cubic threefold since ${}^T X \subseteq \mathbb{P}({}^T A) \cong \mathbb{P}^4$. Moreover, we may twist the Pfaffian representation data to obtain

$${}^T A \subseteq \Lambda^2({}^T V)^\vee$$

and we conclude that ${}^T X$ has a Pfaffian representation over L . By Theorem 3.1, we conclude that ${}^T X$ has an L -point. $\square$

4. CONDITION (A*)

The following theorem can be thought of as a variation of Going-Down.

Theorem 4.1. *Let X and Y be two G -varieties over an algebraically closed field k of characteristic 0, where X is smooth, Y is proper, and G is a finite group. Assume that there exists a rational map $\phi : X \dashrightarrow Y$ which is G -equivariant. If there is a central abelian subgroup A of G , such that X^A contains a G -stable unirational subvariety, then Y^A contains a G -stable unirational subvariety.*

Proof. Let F be a G -stable unirational subvariety of X^A . If ϕ is a morphism, then $\overline{\phi(F)}$ is a G -stable unirational subvariety of Y^A . Thus, it suffices to reduce to the case of a morphism.

From [RY02], there exists a sequence of blowups $Z \rightarrow \cdots \rightarrow X$ with smooth G -stable centers such that the induced map $Z \rightarrow Y$ is a G -equivariant morphism. Thus, it suffices to prove that $(\mathrm{Bl}_C(X))^A$ also contains a G -stable unirational subvariety F' where $\mathrm{Bl}_C(X)$ is the blowup of X at a smooth G -invariant center C .

When $F \not\subseteq C$, we may take F' to be the strict transform of F . When F is contained in C , we obtain an induced action of A on the restriction to F of the normal bundle $\mathcal{N}_{C/X}$. Since A is abelian, we have a character decomposition $(\mathcal{N}_{C/X})|_F = \bigoplus_{\chi} \mathcal{E}_{\chi}$. Pick a non-zero summand $\mathcal{E}_{\chi}$ and set

$$F' = \mathbb{P}(\mathcal{E}_{\chi}) \subseteq \mathbb{P}(\mathcal{N}_{C/X}) \subseteq \mathrm{Bl}_C(X).$$

Then A acts trivially on F' , and F' is unirational since F' is a projective bundle over a unirational variety. By our assumption, G centralizes A . It follows that the summand $\mathcal{E}_{\chi}$ is G -stable and F' is G -stable. $\square$

We are now able to establish that Condition (A*) is necessary for G -weak versality.

Corollary 4.2. *Suppose X is a proper G -variety over an algebraically closed field of characteristic 0, where G is a finite group. If there exists an abelian subgroup A of G such that X^A contains no rational curves and $X^{Z_G(A)}$ is empty, then X is not G -weakly versal.*

Proof. If X is G -weakly versal, then a fortiori it is $Z_G(A)$ -weakly versal. Thus there exists a faithful representation V of $Z_G(A)$ and a rational map $V \dashrightarrow X$. The origin of V satisfies the conditions of Theorem 4.1 for $Z_G(A)$. Now, observe that a $Z_G(A)$ -stable unirational subvariety of X^A either contains a rational curve or is a point. $\square$

5. CUBIC THREEFOLDS

From [CTZ25, Theorem 5.1], we see that a smooth cubic threefold X with a faithful G -action is G -unirational if and only if it satisfies Condition (A), with four possible exceptions:

- (a) $G = C_9 \rtimes C_3$ where X is the Fermat cubic threefold,
- (b) $G = \mathrm{PSL}_2(\mathbb{F}_{11})$ or its subgroup $C_{11} \rtimes C_5$, where X is the Klein cubic threefold,
- (c) $G = \mathfrak{A}_5$ where X is a smooth member of the pencil generated by the Klein and Segre cubic threefolds.

Aside from these exceptions, when Condition (A) holds, the following two strategies suffice to establish G -unirationality:

- If $[G : H] = 2$, then X is G -unirational if and only if X is H -unirational.

- If $X \cap L$ is G -unirational for a G -invariant linear subspace L , then X is G -unirational.

In this section, we settle the remaining four cases.

Theorem 5.1. *The action of $G = C_9 \rtimes C_3$ on the Fermat cubic threefold does not satisfy Condition (A*). In particular, it is not $C_9 \rtimes C_3$ -unirational.*

Proof. There are several actions of groups isomorphic to $C_9 \rtimes C_3$ on the Fermat cubic, some of which are equivariantly unirational. The action singled out in [CTZ25, Remark 5.4] is the following one. Let ζ be a primitive cube root of unity and write

$$X = \{x_1^3 + x_2^3 + x_3^3 + x_4^3 + x_5^3 = 0\} \subset \mathbb{P}^4.$$

The group G is generated in PGL_5 by

$$\begin{aligned} a(x_1, x_2, x_3, x_4, x_5) &= (x_1, x_2, \zeta x_3, \zeta^2 x_4, x_5), \\ b(x_1, x_2, x_3, x_4, x_5) &= (x_1, x_2, x_3, \zeta x_4, \zeta^2 x_5), \\ c(x_1, x_2, x_3, x_4, x_5) &= (x_1, \zeta x_2, x_4, \zeta x_5, x_3). \end{aligned}$$

The center of G is

$$Z = \langle c^3 \rangle \simeq C_3, \quad c^3 = \mathrm{diag}(1, 1, \zeta, \zeta, \zeta).$$

We have that

$$X^Z = E \cup p_1 \cup p_2 \cup p_3,$$

where E is the elliptic curve given by

$$E = \{x_1 = x_2 = x_3^3 + x_4^3 + x_5^3 = 0\},$$

and p_1, p_2, p_3 are three points satisfying

$$p_1 \cup p_2 \cup p_3 = \{x_3 = x_4 = x_5 = x_1^3 + x_2^3 = 0\}.$$

In particular, X^Z contains no rational curves. On the other hand, we can compute that $X^G = \emptyset$. Since $Z_G(Z) = G$, Corollary 4.2 therefore shows that X is not G -weakly versal, and in particular is not G -unirational. $\square$

Theorem 5.2. *The Klein cubic is $\mathrm{PSL}_2(\mathbb{F}_{11})$ -unirational.*

Proof. From [TZ24], we see that the Klein cubic is equivariantly Pfaffian for the order 2 extension

$$1 \rightarrow C_2 \rightarrow \mathrm{SL}_2(\mathbb{F}_{11}) \rightarrow \mathrm{PSL}_2(\mathbb{F}_{11}) \rightarrow 1.$$

The result now follows immediately from Theorem 3.3. $\square$

Remark 5.3. While the methods used to prove the Klein cubic is $\mathrm{PSL}_2(\mathbb{F}_{11})$ -unirational are effective, the resulting explicit map from a linear representation will likely be extremely complicated. I. Cheltsov and C. Shramov have asked if there is a “nice” explicit unirationality construction. The choice of initial linear representation may be important.

Theorem 5.4. *All smooth members of the pencil generated by the Klein and Segre cubics are $\mathfrak{A}_5$ -unirational.*

Proof. Since $\mathfrak{A}_5$ is a subgroup of $\mathrm{PGL}_2(k)$ for k algebraically closed, we may apply Lemma 2.7(e). $\square$

Remark 5.5. A potential alternative proof of Theorem 5.4 follows from the Cassels-Swinnerton-Dyer conjecture for cubic threefolds. In particular, suppose every smooth cubic threefold with an effective 0-cycle of degree 5 has a rational point. Then, a smooth cubic threefold is $\mathfrak{A}_5$ -unirational as soon as it is $\mathfrak{A}_4$ -unirational, which they always are by [CTZ25, Theorem 5.1].

Remark 5.6. It is likely that every smooth member of the pencil is equivariantly Pfaffian for the binary icosahedral group extending $\mathfrak{A}_5$. The result then follows from Theorem 3.3. However, we do not know a conceptual reason for why every smooth member should be equivariantly Pfaffian.

Proof of Theorem 1.5. By Corollary 4.2, Condition (A*) is necessary for G -unirationality. It remains to show that Condition (A*) is sufficient. Note that Condition (A*) implies Condition (A).

If X has isolated singularities, then Condition (A) implies that X is G -unirational by [CTZ25, Theorem 6.1]. If X has non-isolated singularities, then X is G -unirational for every finite group G by [CMTZ26]. If X is smooth, then the three theorems above in conjunction with [CTZ25, Theorem 5.1] show that X is G -unirational whenever it satisfies Condition (A*). $\square$

6. THE PRIME FANO THREEFOLD OF DEGREE 14

Theorem 6.1. *Let $G = \mathrm{PSL}_2(\mathbb{F}_{11})$ and let X be the smooth V_{14} associated with the Klein cubic. Then X is not G -unirational.*

Proof. We apply Corollary 4.2. Let $\sigma \in G$ be an involution and set $A = \langle \sigma \rangle$. There is a unique conjugacy class of involutions in G , and

$$Z_G(A) \simeq \mathfrak{D}_6,$$

where $\mathfrak{D}_6$ has order 12 and $\mathfrak{D}_6/A \simeq \mathfrak{S}_3$. The fixed-locus computation for this action in [TZ24, Section 4] gives

$$X^A = E \cup p \cup q,$$

where E is an elliptic curve, and p and q are two points. Thus X^A contains no rational curves. The same computation records the residual $\mathfrak{S}_3$ -action: it interchanges the two isolated points and has no common fixed point on E . Consequently,

$$X^{Z_G(A)} = (X^A)^{\mathfrak{S}_3} = \emptyset.$$

Corollary 4.2 now shows that X is not G -weakly versal, and hence is not G -unirational. $\square$

The Klein cubic and its associated V_{14} are known not to be $\mathrm{PSL}_2(\mathbb{F}_{11})$ -equivariantly birationally equivalent [BCDP23, Theorem 4.3]. In view of Theorem 5.2 and Theorem 6.1, there is a dominant equivariant rational map $V_{14} \dashrightarrow V_3$, but not conversely. In particular, they are *not* equivariantly

stably birationally equivalent. In contrast, in [TZ24, Proposition 4.1], it was shown that they are “twisted” equivariantly stably birationally equivalent.

AI Declaration. AI assistants were used substantially in this work. The authors feel providing a narrative is of value to other researchers.

The origin of the investigation began with insight of second author, later independently observed by first author, that establishing rationality or unirationality for a class of varieties involved explicit construction of a desired morphism using a set of tools available to algebraic geometers for a hundred years. This should be viewed in contrast to obstructions to rationality or unirationality, which have been the motivation for deep new ideas about algebraic geometry.

The second author directed the first author to the notes [Har] from the 2005 Summer Research Institute in Seattle which provided a list of open questions related to unirationality and rational connectivity. The first author prompted Fable 5 model in the CLI Claude Code harness to inhabit the role of a lab director and review and rank the questions from loc. cit. Once Fable finished, codex ChatGPT Sol 5.6 lab technicians were placed in goal mode with lab orders prepared by Fable for the top three problems.

The technicians made progress on all three questions but the most dramatic was a proof that smooth $(2, 3)$ hypersurfaces in $\mathbb{P}^2 \times \mathbb{P}^2$ are unirational. Inspired by this, the second author then offered the question of $\mathrm{PSL}(2, 7)$ -unirationality of the Klein del Pezzo surfaces which was resolved negatively using the same workflow with the codex technician using what is effectively Condition (A*). When the question of equivariant rationality of V_{14} was posed by the authors, the model used Condition (A*). Like with previous results, the outputs were placed in the repository [Bal].

The investigation of equivariant rationality of the Klein cubic commenced in earnest afterwards but was not complete until the model explicitly had the facts from [Ale] in its context.

Independently the third author elicited Condition (A*) from direct prompting to ChatGPT 6-Astra on the equivariant unirationality of Fermat cubic threefolds.

The authors subsequently reworked these arguments substantially in order to clarify the underlying mathematics, and have verified all of the results. The text was written by the authors, who take full responsibility for the content of this paper.

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